Course notes for

Algebraic Topology I (KSM3E02)

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Day 1 : 4th August, 2026

`quotient topology — homotopy — homotopy equivalence`

Unless otherwise mentioned, throught this course, a space means a topological space and a map $f : X \to Y$ between spaces is always assumed to be continuous.

1.1 A Primer on Quotient Topology

Let us recall few important concepts from point-set topology that crops up in algebraic topology all the time!

Definition 1.1: (*Quotient Topology*)

Given a space X, a set Y, and set map $q : X \to Y$, the *quotient topology* on Y induced by q is the *finest* topology (i.e, the *largest* topology) on Y such that q becomes continuous. We say q is a quotient map.

Explicitly, the quotient topology on Y induced by $q : X \to Y$ is given as

$$\{U \subset Y \mid q^{-1}(U) \subset X \text{ is open}\}.$$

It is easy to verify that the quotient topology as defined above is indeed a topology. In order to understand the definition, note that if we give Y the indiscrete topology then any map $f : X \to Y$ is continuous; on the other hand, if Y has a finer topology, a map into it may fail to be continuous. So, we aim for the finest possible topology on Y. A map $f : X \to Y$ is called a quotient map if the following holds:

$$U \subset Y \text{ is open if and only if } f^{-1}(U) \subset X \text{ is open.}$$

An important property of the quotient topology is the following.

Proposition 1.2: (*Universal Property of the Quotient Topology*)

Suppose $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ are given topological spaces, and $f : X \to Y$ is a set function. Then, the following are equivalent.

1. $\mathcal{T}_Y$ is the quotient topology induced by f.
2. $\mathcal{T}_Y$ is the unique topology having the following property: given any space Z and a set map $g : Y \to Z$, we have g is continuous if and only if $g \circ f$ is continuous.

Quotient topology crops up from equivalence relations. Given an equivalence relation $\sim$ on a space X, denote $X/_\sim$ as the set of equivalence classes. Then, we have a set map $q : X \to X/_\sim$ which maps each element to its equivalence class. We equip $X/_\sim$ with the quotient topology induced by this map. As a consequence, if we have a continuous map $f : X \to Z$ such that $a \sim b \Rightarrow f(a) = f(b)$ (i.e, f respects the equivalence relation), then there is an induced map $\tilde{f} : X/_\sim \to Z$, which is moreover continuous (Proove it!).

Example 1.3: (*Projective Spaces*)

On $X := \mathbb{R}^{n+1} \setminus \{0\}$, consider the equivalence relation: $\mathrm{u} \sim \mathrm{v}$ if and only if $\mathrm{u} = \lambda \mathrm{v}$ for some nonzero scalar $\lambda \in \mathbb{R}$. The induced quotient space, denoted as $\mathbb{RP}^n$, is known as the n^{th} real projective space. Can you see that $\mathbb{RP}^1$ is nothing but the circle S^1?

Let us now recall some constructions that one can perform on topological spaces, all of which are given quotient topology.

1.2 Why Algebraic Topology?!

In point-set topology, we encounter many properties of a space, and then verify which of them are invarient under homeomorphism. Then, this property can be used to distinguish between spaces. As an example, consider the intervals $[0,1]$ and $[0,1)$: they can be homeomorphic since one of them is compact, the other is not, and we know that homeomorphism preserves compactness. Now, if we consider the intervalse $(0,1)$ and $[0,1)$, then compactness is not good enough as none of them are compact. To distinguish them we can use the following trick.

> If possible, suppose $f : [0,1) \to (0,1)$ is a homeomorhpism. Denote $z = f(0)$. Observe that we have an induced map
>
> $$\tilde{f} : (0,1) \to (0,1) \setminus \{z\},$$
>
> which is also a homeomorphism. But this is impossible since deleting a point makes $(0,1)$ disconnected, and we know that connectedness is preserved under homeomorphism.

This trick can be used to distinguish (up to homeomorphism) between $\mathbb{R}$ and $\mathbb{R}^2$ as well. But the trick fails when you try to distinguish between $\mathbb{R}^2$ and $\mathbb{R}^3$.

The point of algebraic topology is to weak the notion of homeomorphism, and yet be able to distinguish between a spaces. Indeed, we shall define the notion of *simply connectedness* and see that $\mathbb{R}^2$ with a point removed is simply connected, but $\mathbb{R}^3$ with a point removed is not.

More generally, we shall try to meaningfully attach algebraic objects (like groups, rings, chain complexes, etc.) to a space, which are (sometimes) computable. Then, using them we shall try to distinguish between spaces. The first step of course is to define the weakened notion of homeomorphism!

1.3 Homotopy and Homotopy Equivalence

Definition 1.4: (*Homotopy*)

Given maps $f, g : X \to Y$, a *homotopy* between them is a map $h : X \times [0,1] \to Y$ such that

$$h(x,0) = f(x), \qquad h(x,1) = g(x), \qquad x \in X.$$

We denote this as $h : f \simeq g$, and say f, g are *homotopic* maps.

Intuitively, a homotopy is a continuous way to deform one map to another.

Proposition 1.5: (*Homtopy is an Equivalence Relation*)

Being homotopic is an equivalence relation on the set $C(X,Y)$ of continuous maps $X \to Y$.

<u>**Proof**</u> **:** We explicitly check the properties.

- **Reflexivity:** Given any $f : X \to Y$, we have a homotopy $h : f \simeq f$ given by $h(x,t) = f(x)$ for $0 \leq t \leq 1$.
- **Symmetry:** Suppose $h : f \simeq g$ is as homotopy. Define the *reversed homotopy* as

$$\begin{aligned} k : X \times [0,1] &\to Y \\ (x,t) &\mapsto h(x, 1-t). \end{aligned}$$

  It is easy to see that $k : g \simeq f$.
- **Transitivity:** Suppose $\varphi : f \simeq g, \psi : g \simeq h$ are homotopies. Then, define the map

$$\begin{aligned} \zeta : X \times [0,1] &\to Y \\ (x,t) &\mapsto \begin{cases} \varphi(x, 2t), & 0 \leq t \leq \frac{1}{2} \\ \psi(x, 2t-1), & \frac{1}{2} \leq t \leq 1. \end{cases} \end{aligned}$$

  Via the *pasting lemma*, it follows that ζ is continuous, and clearly, $\zeta : f \simeq h$ is a homotopy.

Thus, we have $\simeq$ is an equivalence relation on $C(X,Y)$. □

Definition 1.6: (*Homotopy Equivalence*)

A map $f : X \to Y$ is called a *homotopy equivalence* if there is a map $g : Y \to X$ such that $g \circ f \simeq \mathrm{Id}_X$ and $f \circ g \simeq \mathrm{Id}_Y$. We say that X and Y are *homotopy equivalent* as spaces, and denote it as $X \simeq Y$.

Intuitively, homotopy equivalence allows you to deform a space: think of a space as a rubber sheet; you are allowed to squish it, stretch it, but you cannot tear it! Explicitly writing down homotopy equivalence is not always feasible as we shall soon see.

Exercise 1.7: (*Homotopy Equivalence is an Equivalence Relation*)

Show that being homotopy equivalence is an equivalence relation between the class of all topological spaces.

<u>**Hint**</u> **:** Use Exercise 2.1.

Day 2 : 5th August, 2026

`homotopy extension – homotopy lifting`

2.1 Composition and Homtopies

There are two ways to compose homotopies. We can compose them *vertically*

$$\begin{tikzcd}[column sep=huge]
X \arrow[rr, bend left=60, "f"{name=F}] \arrow[rr, "g"{name=G}] \arrow[rr, bend right=60, "h"'{name=Hh}] & & Y
\arrow[Rightarrow, from=F, to=G, "H"']
\arrow[Rightarrow, from=G, to=Hh, "K"']
\arrow[Rightarrow, from=F, to=Hh, bend left, "L"]
\end{tikzcd}$$

This is nothing but the transitivity. We can also compose them *horizontally*

$$\begin{tikzcd}[column sep=huge]
X \arrow[r, bend left, "f_1"{name=F1}] \arrow[r, bend right, "f_2"'{name=F2}] \arrow[rr, bend left=70, "g_1 \circ f_1"{name=T}] \arrow[rr, bend right=70, "g_2 \circ f_2"'{name=B}] & Y \arrow[r, bend left, "g_1"{name=G1}] \arrow[r, bend right, "g_2"'{name=G2}] & Z
\arrow[Rightarrow, from=F1, to=F2, "H"']
\arrow[Rightarrow, from=G1, to=G2, "K"']
\arrow[Rightarrow, from=T, to=B, bend left, "L"]
\end{tikzcd}$$

Exercise 2.1:

Construct the horizontal composition of two homotopies.

2.2 Retraction

Definition 2.2: (*Retraction*)

Given a subspace $A \subset X$, a *retraction* is a map $r : X \to A$ such that $r \circ \iota = \mathrm{Id}_A$, where $\iota : A \hookrightarrow X$ is the inclusion. We say A is a *retract* of X. In other words, we have a commuting diagram

$$\begin{tikzcd}
A \arrow[r, hook, "\iota"'] & X \arrow[l, bend right, "r"']
\end{tikzcd}$$

As an example, any point in a space is a retract. We shall see that the boundary circle is *not* a retract of the closed disc (and more generally, boundary sphere is not a retract of the closed ball). Try to justify that $A = \{0, 1\} \subset [0, 1]$ is *not* a retract of $[0, 1]$!

Exercise 2.3:

Suppose X is a T_2-space (i.e, Hausdorff). If $A \subset X$ is a retract, verify that A must be a closed subset.

Hint : For a T_2-space X, if we have two maps $f, g : X \to Y$, then the diagonal $\Delta = \{x \in X \mid f(x) = g(x)\}$ is a closed subset.

Let us now specialize homotopy equivalence to the case of retraction, which is easier to visualize.

Definition 2.4: (*Deformation Retract*)

A subspace $A \subset X$ is a *deformation retract* if there is a retraction $r : X \to A$ satisfying $r \circ \iota = \mathrm{Id}_A$, and there is a homotopy $\iota \circ r \simeq \mathrm{Id}_X$. Explicitly, we need a homotopy $H : X \times [0, 1] \to X$ such that

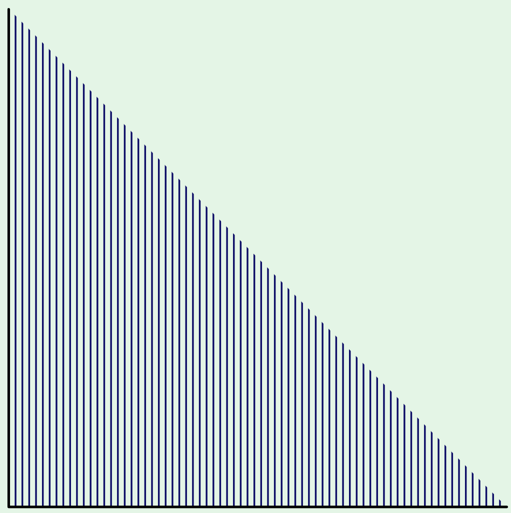
$$H(x, 1) = x, \quad x \in X, \qquad H(x, 0) \in A, \quad x \in X, \qquad H(a, 0) = a, \quad a \in A.$$

We say $A \subset X$ is a *strong deformation retract* if moreover the homotopy restricts to identity on A, i.e, if $H(a, t) = a$ for all $a \in A$ and $0 \le t \le 1$.

It is clear from the definition that a deformation retract is always a homotopy equivalence. There are (pathological) examples which are deformation retracts, but *not* strongly.

Example 2.5:

Consider the following comb-like space.



Explicitly, the space X can be identified as the following subspace of $\mathbb{R}^2$

$$X = ([0,1] \times \{0\}) \cup \bigcup_{r \in [0,1] \cap \mathbb{Q}} \{r\} \times [0, 1-r]$$

That is, keep the horizontal interval $[0,1] \times \{0\}$, and then for each rational point r on it add the vertical line of length $1 - r$.

We claim that X strongly deformation retrats onto any point on the horizontal line. Indeed, it is easy to construct a linear homtopy which pushes everything down to the horizontal line, and then collapses the line to a designated point; this designated point is never moved throughout.

On the other hand, for any point *not* on the horizontal line, there is always a deformation retract: just perform the previous deformation to the point, say, $(0,0)$ and then slide this point to any other point, speeding up as necessary. But for such a point, there cannot be a strong deformation retract. To see this intuitively, observe that if there is a strong deformation retract to a point, then in any neighborhood of that point must contain a smaller neighborhood which also collapses to that same point. But this is clearly not the case since you can always find a neighborhood of such a point which cannot be path connected (and hence cannot collapse to a point).

Let us give a name to spaces which are homotopy equivalent to a point.

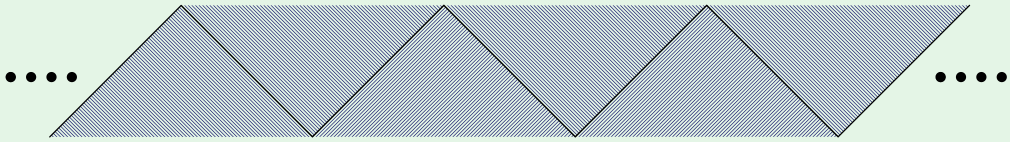
Definition 2.6: (*Contractible Space*)

A space X is said to be *contracitble* if it is homotopy equivalent to a point. We denote it as $X \simeq \star$.

We have seen that any space admits a retraction to any point in it. It is easy to see that if a space deformation retracts to a point, then it is contractible, and vice versa. This does not hold true for *strong* deformation retract.

Example 2.7:

This example builds upon Example 2.5. Consider the following subspace X of $\mathbb{R}^2$



This is obtained from the comb-like space in Example 2.5 by arranging them in the infinite zigzag pattern!

We claim that this space X is *contractible*, but does not *strongly* deformation retract to any point. Intuitively, a strong deformation retract to a point x_0 will imply that there for any neighborhood of x_0, there is a smaller neighborhood which contracts to x_0: this is clearly not the case here (do a case-by-case analysis of different kinds of points).

In order to see why this space is contractible, we do the following. Firstly, for each point x in the space, we have a unit-speed path $r_x : [0, \infty) \to X$ which follows the *bristle* of the comb where x belongs till it hits the bold zigzag line, and then moves along the zigzag to the right. Clearly if x is already on the zigzag, it just starts following it. Consider the following map

$$\begin{aligned} F : X \times [0,1] &\to X \\ (x,t) &\mapsto r_x(t). \end{aligned}$$

One can check that F is a continuous map, and hence a homotopy between the Id_X to a map which ends in the zigzag. Finally, the zigzag is just a copy of $\mathbb{R}$, and hence contractible. Thus, composing we can get a contracting homotopy. In other words, $X \simeq \star$.

Caution 2.8:

According to Hatcher, a "deformation retract" is precisely what we call a *strong* deformation retract. Hatcher also defines a notion of "deformation retract in the weak sense": this asks for a deformation retraction $H : X \times [0,1] \to X$ such that $H(a,t) \in A$ for all $a \in A$ and for all time $0 \leq t \leq 1$.

2.3 Homotopy Extension Property

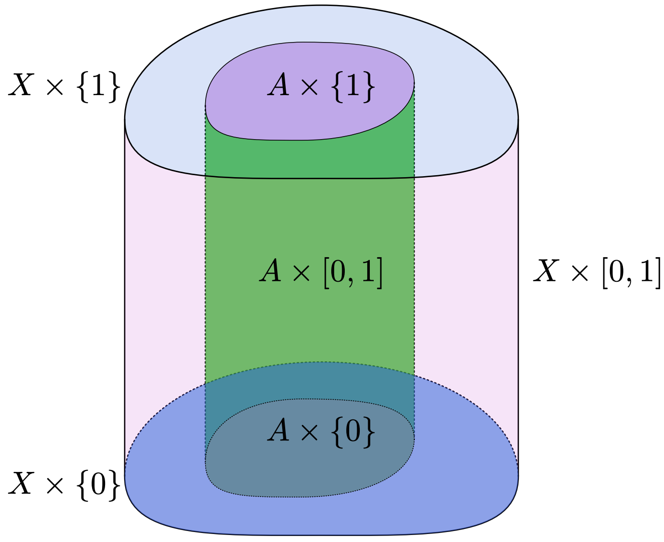
Suppose $A \subset X$ is a subspace. Assume that we are given a map $f : X \to Y$, and a homotopy $h : A \times [0,1] \to Y$ from $f|_A : A \to Y$. We say, the pair (X, A) has *homotopy extension property* (or *HEP*) with respect to the space Y if there exists a homotopy $H : X \times [0,1] \to Y$ from f, which extends the homotopy h. That is,

$$H(x,0) = f(x) \quad x \in X, \qquad H(a,t) = h(a,t) \quad a \in A, 0 \leq t \leq 1.$$

The schematic picture explains what is going on. We are given a continuous map on the subspace

$$A \times [0,1] \cup X \times \{0\} \subset X \times [0,1],$$

and we wish to extend it to the wholse space $X \times [0,1]$.



Homotopy extension for $A \subset X$.

Definition 2.9: (*Cofibration*)

The inclusion map $\iota : A \hookrightarrow X$ is called a *(Hurewicz) cofibration* if it has homotopy extension property with respect to every space Y.

Remark 2.10:

More generally, we can define HEP for any map $f : A \to X$, not just inclusion maps. Indeed, this can be formulated via the following diagram

$$\begin{tikzcd} A \arrow[r, "h"] \arrow[d, "f"'] & Y^{[0,1]} \arrow[d, "e"] \\ X \arrow[r, "g"'] \arrow[ur, dashed, "H"] & Y \end{tikzcd}$$

Here, $Y^{[0,1]}$ denotes the space of continuous maps $[0,1] \to Y$, and $e : Y^{[0,1]} \to Y$ is evaluation at 0. A homotopy $h : A \times [0,1] \to Y$ can be interprated as a map $h : A \to Y^{[0,1]}$. The map $f : A \to X$ is said to have HEP agains the space Y if for any given map $g : X \to Y$ and a homotopy $h : A \times [0,1] \to Y$, such that $h|_{A\times\{0\}} = g \circ f$, there exists a homotopy $H : X \times [0,1] \to Y$ which makes the diagram commutative. One then defines $f : A \to X$ to be a *(Hurewicz) cofibration* if it has HEP agains all space Y.

Turns out any cofibration $f : A \to X$ is actually injective, and moreover, f is a homeomorphism onto its image. Thus, we are not losing any generality with our definition.

Let us now see a prototypical example of a cofibration!

Example 2.11: ($S^{n-1} \hookrightarrow D^n$ *is a Cofibration*)

Consider the inclusion $\iota : S^{n-1} \hookrightarrow D^n$ of the unit $(n-1)$-sphere S^{n-1} as the boundary of the closed unit disc D^n in $\mathbb{R}^n$. We claim that this has HEP agains any space, and hence, is a cofibration. Suppose $f : D^n \to Y$ is a map, and $h : S^{n-1} \times [0,1] \to Y$ is a homotopy of $f|_{S^{n-1}}$. Consider the subspace

$$X := S^{n-1} \times [0,1] \cup D^n \times \{0\} \subset D^n \times [0,1].$$

Clearly, we are given a map $F : X \to Y$, which we intend to extend to $D^n \times [0,1]$. One can imagine X as a hollowed out cylinder (with the bottom intact, basically like a bucket). One can define a *retraction* $r : D^n \times [0,1] \to X$ as follows. Consider the point $z = \mathbf{0} \times \{2\}$, the center of the disc $D^n \times \{2\}$. Given any point p in the solid cylinder $D^n \times [0,1]$, draw a straight line from z which passes throught p, and then record the (unique) point q where this line hits the bucket X. Define $r : D^n \times [0,1] \to X$ which maps $p \mapsto q$. It is easy to see that r is continuous, and moreover, r is identity when restricted to the subspace X. Define the homotopy $D^n \times [0,1] \to Y$ as the composition $D^n \times [0,1] \xrightarrow{r} X \xrightarrow{F} Y$. Since r is identity on X, this composition is indeed an extension of F. Thus, $\iota : S^{n-1} \hookrightarrow D^n$ is a cofibration.

Exercise 2.12:

The construction of retraction map in Example 2.11 is also known as the *orthographic projection.* Figure out an explicit formula for the map!

Day 3 : 6$^{\text{th}}$ August, 2026

`contractible spaces — disjoint union — attaching spaces — mapping cylinder — mapping cone`

3.1 A Detour into Contractible Spaces

We have already defined contractible sapces (Definition 2.6). In Example 2.7 we saw that contractible spaces need to *strongly* deformation retract onto any point.

Exercise 3.1:

Given a space X, verify that the following are equivalent.

1. X is contractible.
2. X admits a deformation retract to some $x_0 \in X$.

Next, let us see some nontrivial but important examples of contractible spaces.

Example 3.2: (*Path Space is contractible*)

Given a space X, fix a point $x_0 \in X$. Consider the space of all paths in X starting at x_0

$$\mathcal{P} = \mathcal{P}(X, x_0) = \{\gamma : [0,1] \to X \mid \gamma(0) = x_0\}.$$

Give it the compact-open topology. Recall: the compact-open topology on the space $C(X, Y)$ of continuous maps is generated by the sets

$$\{f : X \to Y \mid f(C) \subset U\}, \quad C \underset{\text{compact}}{\subset} X, \quad U \underset{\text{open}}{\subset} Y.$$

Sinse $[0,1]$ is locally compact, it follows that the evaluation map

$$\begin{aligned} e : \mathcal{P} &\to X \\ \gamma &\mapsto \gamma(1) \end{aligned}$$

is continuous. We claim that the path space $\mathcal{P}$ is contractible.

Intuitively, the contracing homotopy is given by contracting every path back to the origin x_0, like sucking a bunch of spaghetti! Explicitly, we can consider the map

$$\begin{aligned} H : \mathcal{P} \times [0,1] &\to \mathcal{P} \\ (\gamma, t) &\mapsto (s \mapsto \gamma((1-t)s)). \end{aligned}$$

Clearly, we have $H|_{t=0} = \mathrm{Id}_{\mathcal{P}}$, and $H|_{t=1}$ maps every path γ to the constant path $\gamma_0 : s \mapsto x_0$. Thus, H is a strong deformation retract of $\mathcal{P}$ to the *point* γ_0. The fact that H is continuous is a consequence of the compact-open topology.

Example 3.3: (S^∞ *is contractible*)

Recall that the 0^{th} sphere S^0 is nothing but two points, clearly it is not even connected. We know S^1 is path connected, and later see that it is not *simply connected,* in particular S^1 is not contractible. Similarly, using other machinary (namely, either homology or higher homotopy groups) one can show that the n -sphere S^n is not contractible either.

Let us define the inifinite-dimensional sphere S^∞. There are many ways to do so. Recall we have the following inclusions

$$S^0 \hookrightarrow S^1 \hookrightarrow S^2 \hookrightarrow S^3 \hookrightarrow \cdots,$$

where the inclusions are the middle sphere. One defines S^∞ as the colimit of these spaces. Without going into this abstract construction, let us define it in another way.

Consider the space X to be the collection of sequences $\mathrm{x} = (x_0, x_1, x_2, \ldots)$ of real numbers, such that all but finitely many of them are 0 (i.e, for some large enough K, we have $x_n = 0$ for $n \geq K$). This space X is a normed linear space via the ℓ^2-norm

$$\|\mathrm{x}\| = \sqrt{\sum x_i^2}.$$

Note that the sum is a finite sum. One then defines

$$S^\infty = \{\mathrm{x} \in X \mid \|\mathrm{x}\| = 1\}$$

as the unit norm sphere in X.

We claim that S^∞ is contractible. In order to see this, consider the shift map $T : X \to X$ defined as

$$T(x_0, x_1, \ldots) = (0, x_0, x_1, \ldots).$$

One can check that T is continuous and injective. Note that for any $\mathrm{x} \in S^\infty$, we have $T(\mathrm{x})$ is *not* a multiple of x. In particular, the line $t\mathrm{x} + (1-t)T(\mathrm{x})$ is never 0. Consequently, we have a continuous map

$$\begin{aligned} F : S^\infty \times [0,1] &\to S^\infty \\ (\mathrm{x}, t) &\mapsto \frac{t\mathrm{x} + (1-t)T(\mathrm{x})}{\|t\mathrm{x} + (1-t)T(\mathrm{x})\|}. \end{aligned}$$

This is a homotopy between Id and the normalized shift map $f : \mathrm{x} \mapsto \frac{T(\mathrm{x})}{\|T(\mathrm{x})\|}$. Next, note that the point $x_0 = (1, 0, 0, \ldots) \in S^\infty$ is *not* in the image of this normalized shift map. Consequently, the line joining x_0 with any point in the image of f does not pass through the origin, and we have another homotopy

$$\begin{aligned} G : S^\infty \times [0,1] &\to S^\infty \\ (x, t) &\mapsto \frac{tx_0 + (1-t)f(\mathrm{x})}{\|tx_0 + (1-t)f(\mathrm{x})\|}. \end{aligned}$$

This is a homotopy from f to the constant map at x_0. Composing the two homotopies, we get a deformation retract of S^∞ to the point x_0. In particular, S^∞ is contractible.

3.2 Disjoint Union

Let us now define a few important constructions in algebraic topology, which are all special instances of quotient spaces.

Example 3.4: (*Disjoint Union*)

Given two spaces X, Y, consider the *disjoint union* $X \sqcup Y$ as a set. Explicitly, you can consider

$$X \sqcup Y := \{(x, X) \mid x \in X\} \cup \{(y, Y) \mid y \in Y\},$$

and then you can identify X, Y as *subsets* of $X \sqcup Y$ in the obvious way. Next, give it the following topology:

$$U \subset X \sqcup Y \text{ is open if and only if both } U \cap X \text{ and } U \cap Y \text{ are open.}$$

Verify that this topology is induced by the following equivalence relation:

$$(u, A) \sim (v, B) \text{ if and only if } (u, A) = (v, B).$$

$X \sqcup Y$ is also known as the *topological sum* of the spaces.

The above description of the disjoint union topology is very cumbersome. The point is that given two subsets $A, B \subset C$, you can construct the union $A \cup B \subset C$, if there are elements repeating in both A and B they are not considered as distinct elements. On the other hand, for disjoint union, you want them to be distinct! That is why, any element $a \in A$ is *tagged* by a unique ID, say, (a, A) and similarly, $b \in B$ is tagged as (b, B). As an explicit example, consider $A = \{1, 2\}, B = \{2, 3\}$, then $A \cup B$ has three elements, namely $\{1, 2, 3\}$, whereas $A \sqcup B$ has 4 elements: $\{(1, A), (2, A), (2, B), (3, B)\}$.

Exercise 3.5:

Consider the spaces $A = [0, \infty)$ and $B = (-\infty, 0]$. Clearly $A \cup B = \mathbb{R}$. Justify that $A \sqcup B$ is *not* homeomorphic to $\mathbb{R}$.

Hint : What are the connected components of $A \sqcup B$?

Exercise 3.6:

Consider $A = \mathbb{Q}$ and $B = \mathbb{R} \setminus \mathbb{Q}$ as subspaces of $\mathbb{R}$. Is $A \sqcup B$ homeomorphic to $\mathbb{R}$?

The concept of disjoint union easily generalizes to arbitrary collection of spaces. Indeed, given spaces $\{X_\alpha\}_{\alpha \in \Lambda}$, we have the disjoint union $\bigsqcup_{\alpha \in \Lambda} X_\alpha$, where the topology is given as $U \subset \bigsqcup_{\alpha \in \Lambda} X_\alpha$ is open if and only if $U \cap X_\alpha$ is open for each $\alpha \in \Lambda$.

Exercise 3.7: (*Universal Property of Disjoint Union*)

Verify that given any space Z and any continuous maps $f : X \to Z, g : Y \to Z$, there exists a *unique* continuous map $h : X \sqcup Y \to Z$ such that $h|_X = f$ and $h|_Y = g$. Moreover, this property can be used to define the disjoint union:

> if there exists a space W such that for any $f : X \to Z, g : Y \to Z$ there is a unique map $h : W \to Z$ satisfying $h|_X = f, h|_Y = g$, then W is (uniquely) homeomorphic to $X \sqcup Y$.

One denotes, $f \sqcup g : X \sqcup Y \to Z$. Do this exercise for an arbitrary collection of spaces.

3.3 Attaching Spaces

Suppose $A \subset X$ is a subspace, and $f : A \to Y$ is a map. We would like to *attach* (or glue) X to Y along this map.

Definition 3.8: (*Attaching Space*)

Given a subspace $A \subset X$ and a map $f : A \to Y$, the *attaching space* $X \cup_f Y$ is defined as the quotient space induced by the following equivallence relation on the disjoint union $X \sqcup Y$: $u \sim v$ if and only if one of the following holds

i. $u = v$.

ii. $u, v \in A$ and $f(u) = f(v)$.

iii. $u \in A$ and $v = f(u)$.

iv. $v \in A$ and $u = f(v)$

The map f is called the *attaching map*.

At first glance, this equivalence relation might seem hard to parse. Instead, we look at the universal property for the attaching spaces. Firstly, note that we have two canonical maps $p : X \to X \cup_f Y, q : Y \to X \cup_f Y$, both of which are continuous. Given an arbitrary space Z, suppose we have maps $\varphi : X \to Z, \psi : Y \to Z$. Then, as discussed earlier, we have a unique map $\varphi \sqcup \psi : X \sqcup Y \to Z$. Now, given a subspace $A \subset X$, and a map $f : A \to Z$, suppose moreover that $\varphi|_A = \psi \circ f$. Then, there exists a *unique* map $h : X \cup_f Y \to Z$ such that $h \circ p = \varphi, h \circ q = \psi$. That is, we have the diagram

$$\begin{tikzcd}
A \arrow[r, hook] \arrow[d, "f"'] & X \arrow[d, "p"] \arrow[ddr, bend left, "\varphi"] & \\
Y \arrow[r, "q"'] \arrow[drr, bend right, "\psi"'] & X \cup_f Y \arrow[dr, dashed, "\exists! h"] & \\
& & Z
\end{tikzcd}$$

Here is an example, we shall see more soon.

Example 3.9:

- X, Y be the closed unit disc in $\mathbb{R}^2$, A be the unit circle, and $f : A \to Y$ be the inclusion map. Then, $X \cup_f Y$ is homeomorphic to the 2-sphere S^2.
- X be the closed unit disc in $\mathbb{R}^2$, Y be the space with a single point, A be the unit circle and $f : A \to Y$ be the (obvious) constant map. Then, $X \cup_f Y$ is again homeomorphic to S^2.

The last example is also known as an *identification space*.

Definition 3.10: (*Identification Space*)

Given a subspace $A \subset X$, the *identification space* X/A is defined as the attaching space $X \cup_f \{\star\}$, where $f : A \to \{\star\}$ is the constant map. Alternatively, X/A is defined as the quotient space induced from the following equivalence relation on X:

$$u \sim v \text{ if and only if either } u, v \in A, \text{ or } u \notin A, v \notin A, u = v.$$

Intuitively, the identification space collapses the subspace $A \subset X$, and identifies it to a single point.

Exercise 3.11:

Given X as the closed unit ball in $\mathbb{R}^n$ and $A \subset X$ as the closed unit sphere S^{n-1}, verify that X/A is homeomorphic to the n-sphere S^n.

Exercise 3.12:

Suppose $A, B \subset X$ are subspaces such that $X = \mathring{A} \cup \mathring{B}$, i.e, the interiors of A and B covers X. Show that the attaching space $A \cup B$ along the intersection $A \cap B$ is homeomorphic to X.

3.4 Mapping Cylinder

The concept of attaching spaces gives rise to many important constructions in algebraic topology.

Definition 3.13: (*Mapping Cylinder*)

Given $f : X \to Y$ a map, the *mapping cylinder* is constructed as the attaching space $M_f := (X \times [0,1]) \cup_f Y$, where we have identified f as the attaching map $f : X \times \{0\} \to Y$. We have an inclusion map

$$\begin{aligned} \iota : X &\hookrightarrow M_f \\ x &\mapsto [(x,1)], \end{aligned}$$

and a retraction map

$$\begin{aligned} r : M_f &\to Y \\ [(x,t)] &\mapsto f(x) \\ [y] &\mapsto y. \end{aligned}$$

Proposition 3.14: (*Mapping Cylinder and Cofibration*)

Given any $f : X \to Y$, the inclusion $\iota : X \hookrightarrow M_f$ in the mapping cylinder is a cofibration, and $r : M_f \to Y$ is a homotopy equivalence. In fact, r is a strong deformation retract.

This construction gives us the following commutative diagram

$$\begin{tikzcd} & M_f \arrow[dr, bend left, two heads, "r"', "s.d.r"] & \\ X \arrow[ur, bend left, hook, "\iota"] \arrow[rr, "f"'] & & Y \end{tikzcd}$$

One says that any map $f : X \to Y$ can be replaced (up to homotopy) by a cofibration $\iota : X \to M_f$ into the mapping cylinder.

3.5 Homotopy Lifting Property

The dual concept of homotopy *extension* is homotopy *lifitng.* Given a map $p : E \to B$, we say p has *homotopy lifting property* (or *HLP*) agains a spaces X if given any map $f : X \to E$ and any homotopy $h : X \times [0,1] \to B$ satisfying $h|_{t=0} = p \circ f$, there exists a lift $H : X \times [0,1] \to E$ such that

$$H(x,0) = f(x) \quad x \in X, \qquad p \circ H(x,t) = h(x,t) \quad x \in X, 0 \le t \le 1.$$

In other words, given a commutative diagram

$$\begin{tikzcd} X \times \{0\} \arrow[r, "f"] \arrow[d, hook] & E \arrow[d, "p"] \\ X \times [0,1] \arrow[r, "h"'] \arrow[ur, dashed, "H"] & B \end{tikzcd}$$

there exists a lift H making everything commute.

Definition 3.15: (*Fibration*)

A map $p : E \to B$ is called a *(Hurewicz) fibration* if it has homotopy lifting property against any space X.

Let us now see few examples of fibrations.

Example 3.16:

Given any space X, we have the constant map $p : X \to \{\star\}$, and it is easy to see a fibration: the lift is given by constant homotopy. More generally, given a product $B \times F$, consider the projection $\pi : B \times F \to B$. Then, π is a fibration. Indeed, given any diagram

$$\begin{tikzcd} X \times \{0\} \arrow[r, "f"] \arrow[d, hook] & B \times F \arrow[d, "p"] \\ X \times [0,1] \arrow[r, "h"'] & B \end{tikzcd}$$

define $H(x,t) = f(x)$. It is easy to see that H is a homotopy lifting here. More generally, one can show that a fiber bundle (which is a *twisted* product space) is a fibration if B is a paracompact space.

Remark 3.17:

A more general class of fibration is given by Serre fibrations: a map $p : E \to B$ is a *Serre fibration* if it has HLP against all CW complexed. Equivalently, one can define Serre fibrations as those maps $p : E \to B$ which has HLP for all diagrams of the form

$$\begin{tikzcd}
S^{n-1} \times \{0\} \arrow[r, "f"] \arrow[d, hook] & B \times F \arrow[d, "p"] \\
X \times [0,1] \arrow[r, "h"'] & B
\end{tikzcd}$$